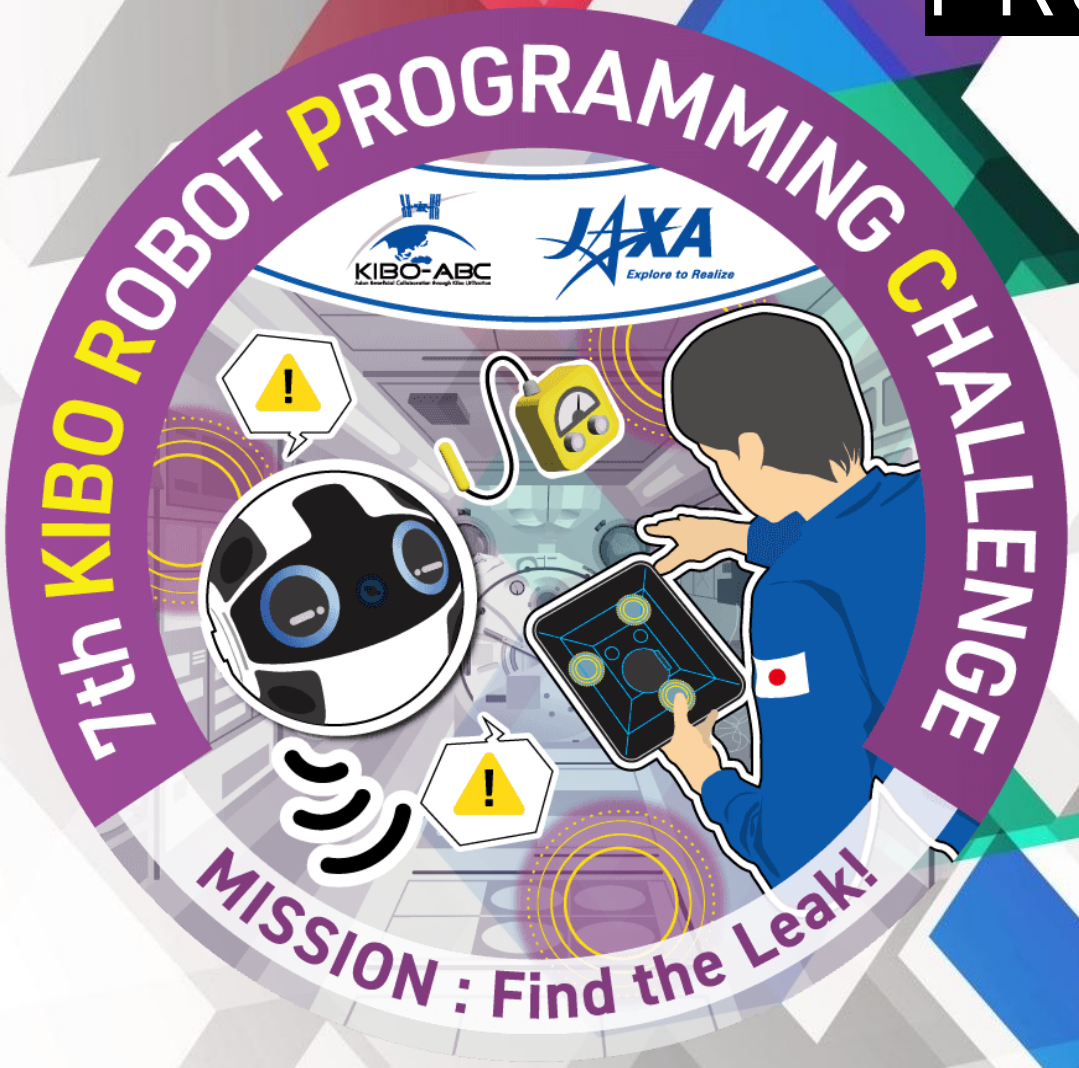
THE 7TH KIBO ROBOT PROGRAMMING CHALLENGE GUIDANCE SESSION 2



October 21, 2026

Kibo-RPC Secretariat

AGENDA

1. **Overview of the Kibo Robot Programming Challenge (Kibo-RPC)**
2. **Outline of the 7th Kibo-RPC**
3. **Preliminary Round**
4. **ISS Final Round Run**
5. **Beginner Course**
6. **Q&A**

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OVERVIEW OF THE KIBO-RPC

- The Kibo Robot Programming Challenge is a robot programming competition in which robots are controlled by student programs on the International Space Station (ISS).
- The program is open for students from elementary school to graduate school[*1] from all over the world (mainly in the Asia-Pacific region)[*2] .

[*1] Some exceptions are available, so please check with the local organizers in charge.

[*2] Please refer to the guidebook for details.

- The program allows students to:
  - Gain scientific, technical and mathematical skills and expertise.
  - Learn about position and attitude estimation for space robots, the need for corrective actions, and how to achieve them.
  - Route design and team strategy

Please refer to the following website for reference:

https://jaxa.krpc.jp/

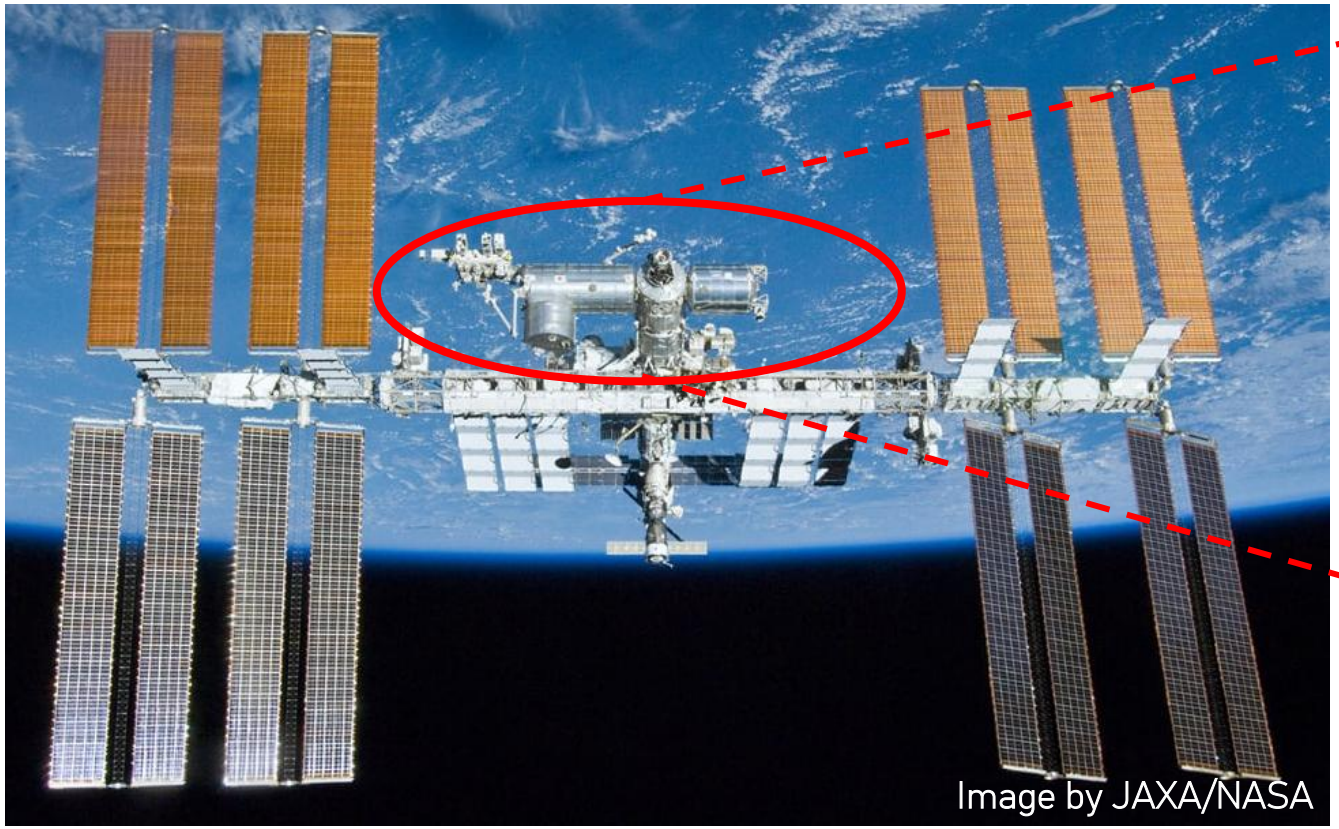


Image by JAXA/NASA

OVERVIEW OF THE KIBO-RPC

- The International Space Station (ISS) is a huge crewed experimental facility built about 400 km above the ground.

Overall view of the ISS



Kibo Japanese Experiment Module

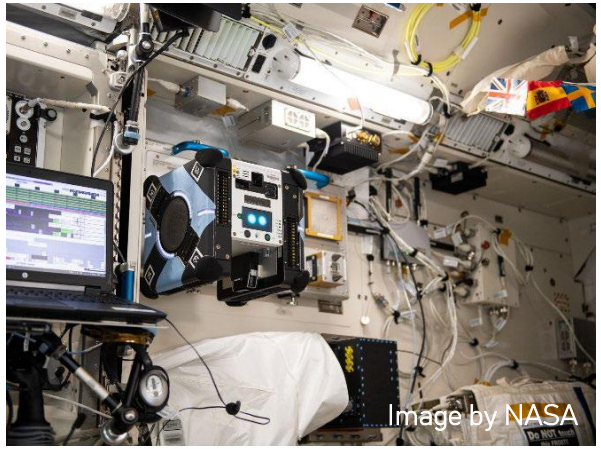


Kibo-RPC will be conducted in the Japanese Experiment Module Kibo.

OVERVIEW OF THE KIBO-RPC

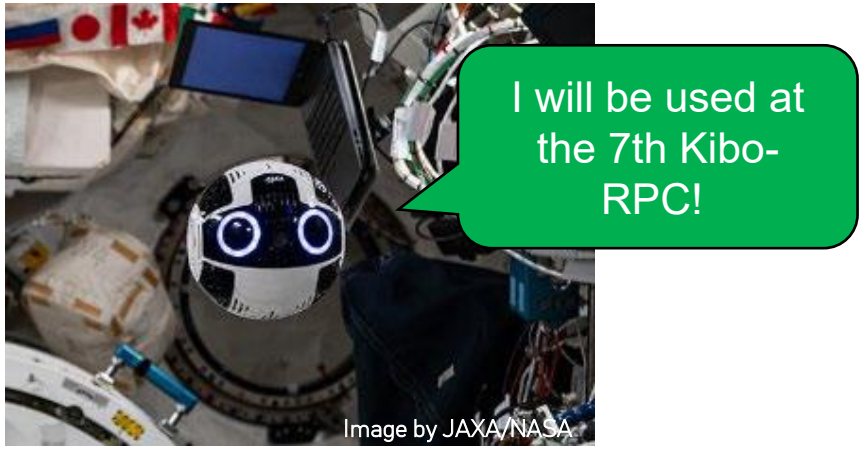
- Introduction of free-flying robots on board the ISS

Astrobee



Astrobee is NASA's free-flying robotic system that will help astronauts reduce the time they spend on routine duties, leaving them to focus more on the things that only humans can do.
Working autonomously or via remote control by astronauts, flight controllers, or researchers on the ground, the robots can perform tasks such as taking inventory, documenting experiments, or moving small items or cargo throughout the station.

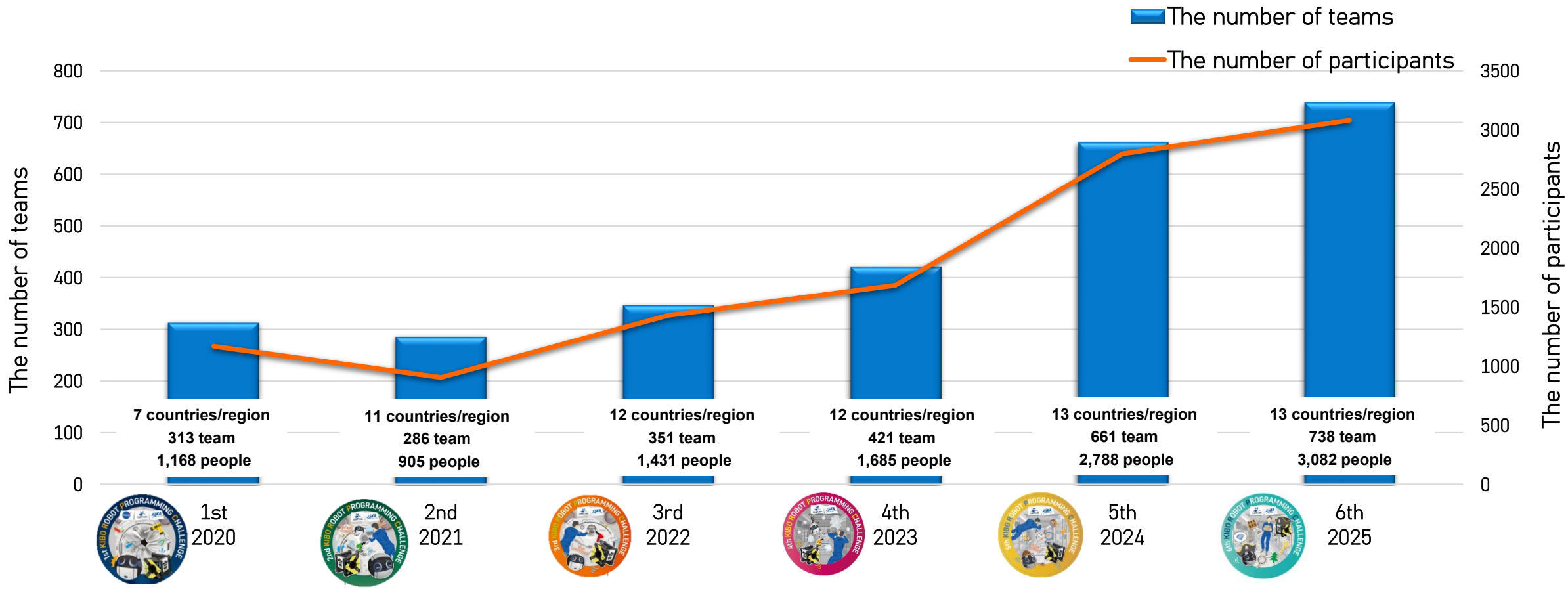
Int-Ball2



Int-Ball2 is a successor of Int-Ball, free-flying camera drone designed to ultimately eliminate the need for crew members to capture routine video footage aboard the ISS/Kibo.
Similar to current consumer-grade cameras, Int-Ball2 is used by onboard crew to provide flexible views for ground operators.

OVERVIEW OF THE KIBO-RPC

- Introduction of Kibo-RPC to date

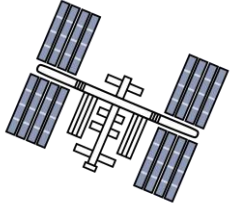


1st 2020	2nd 2021	3rd 2022	4th 2023	5th 2024	6th 2025
Mission: Create a program to automatically control the Astrobee and repair Kibo's damaged parts	**Mission:** Do a complete repair and report completion of the mission to the astronauts	**Mission:** Repair the air leaks that were again caused by the space debris impact	**Mission:** Repair the leakage points by laser irradiation of Astrobee	**Mission:** Work with Astrobee to find the items the astronaut is looking for!	**Mission:** Work with Astrobee to find hidden treasure!

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OUTLINE OF THE 7TH KIBO-RPC



Mission: Crisis on the ISS. Track down the invisible anomaly

A very slight drop in atmospheric pressure was detected in the system monitoring logs of the International Space Station's Japanese Experiment Module 'Kibo'. While this level of pressure change could be due to an equipment error, if it indicates an air leak, careful investigation is necessary.
The cause has not yet been identified, and it has not been confirmed whether or not there is a leak. At this stage, careful analysis is needed before astronauts can conduct a direct investigation. Therefore, a preliminary inspection will be conducted by operating the Int-Ball2 robot flying inside the ISS.

Based on past operational data and case studies, potential leak locations include joints, hatch areas, and the ends of experimental racks. Check Points with AR markers have been installed at each of these potential locations.
Int-Ball2 will visit all Check Points to collect inspection data. However, leaks are so minute that they are invisible to the naked eye. For inspection cameras and sensors to diagnose correctly, measurements must be taken accurately at the specified positions and attitudes. The more accurately you reach the target point, the more accurately you can determine the presence and severity of any abnormalities.
Once all potential locations have been surveyed, the data collected will be integrated and analyzed. This may reveal the true leak location, or it may turn out to be a false alarm. We will try to identify the cause based on the limited information available.

Precisely control Int-Ball2 and uncover the true nature of the anomaly.

*This story is fictional.



Mission and Rules

Regular Course

OUTLINE OF THE 7TH KIBO-RPC

Overview of the Game (Mission)

① The game starts when Int-Ball2 automatically undocks from the Dock Station.

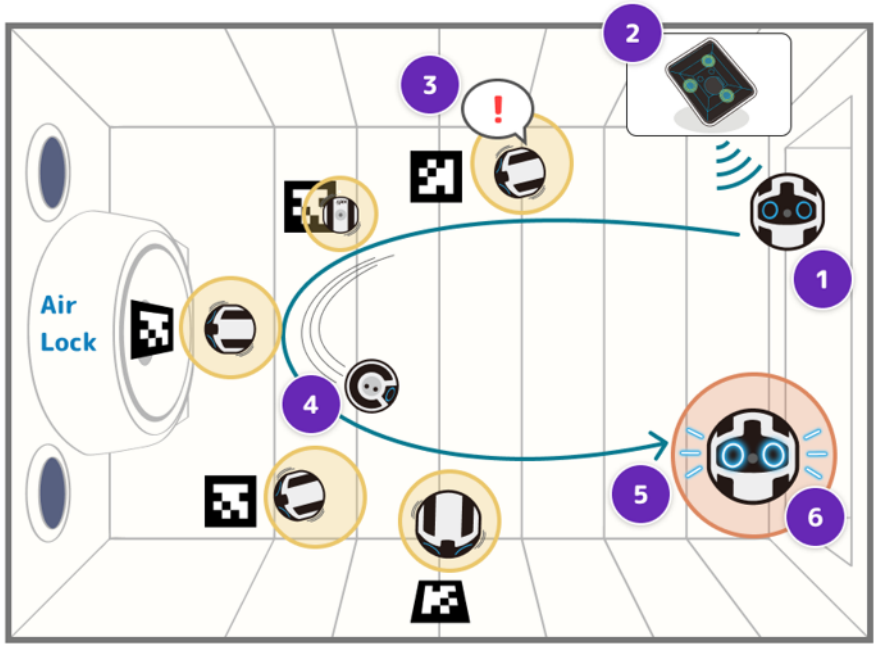
② Read the automatically provided position and attitude information of the Check Points.

③ Visit the provided Check Points using an efficient route.

④ At each Check Point, read the AR marker while paying attention to the required position and attitude.

⑤ The mission is complete when Int-Ball2 has visited all Check Points and reached the Goal Check Point.

⑥ Submit the mission completion report.



OUTLINE OF THE 7TH KIBO-RPC

① **Mission Start**: Moment when the mission begins when the undocking operation is completed. The mission timer for the time limit starts at this point.

② **Mission Finish**: Moment when wither the mission time limit expires or the mission completion report is initiated. Any actions performed after Mission Finish are not eligible for scoring.

③ **AR marker**: Marker used as the origin for recognizing the relative position and attitude of Int-Ball2 and for generating Check Points. There are 12 AR markers in the Kibo module.

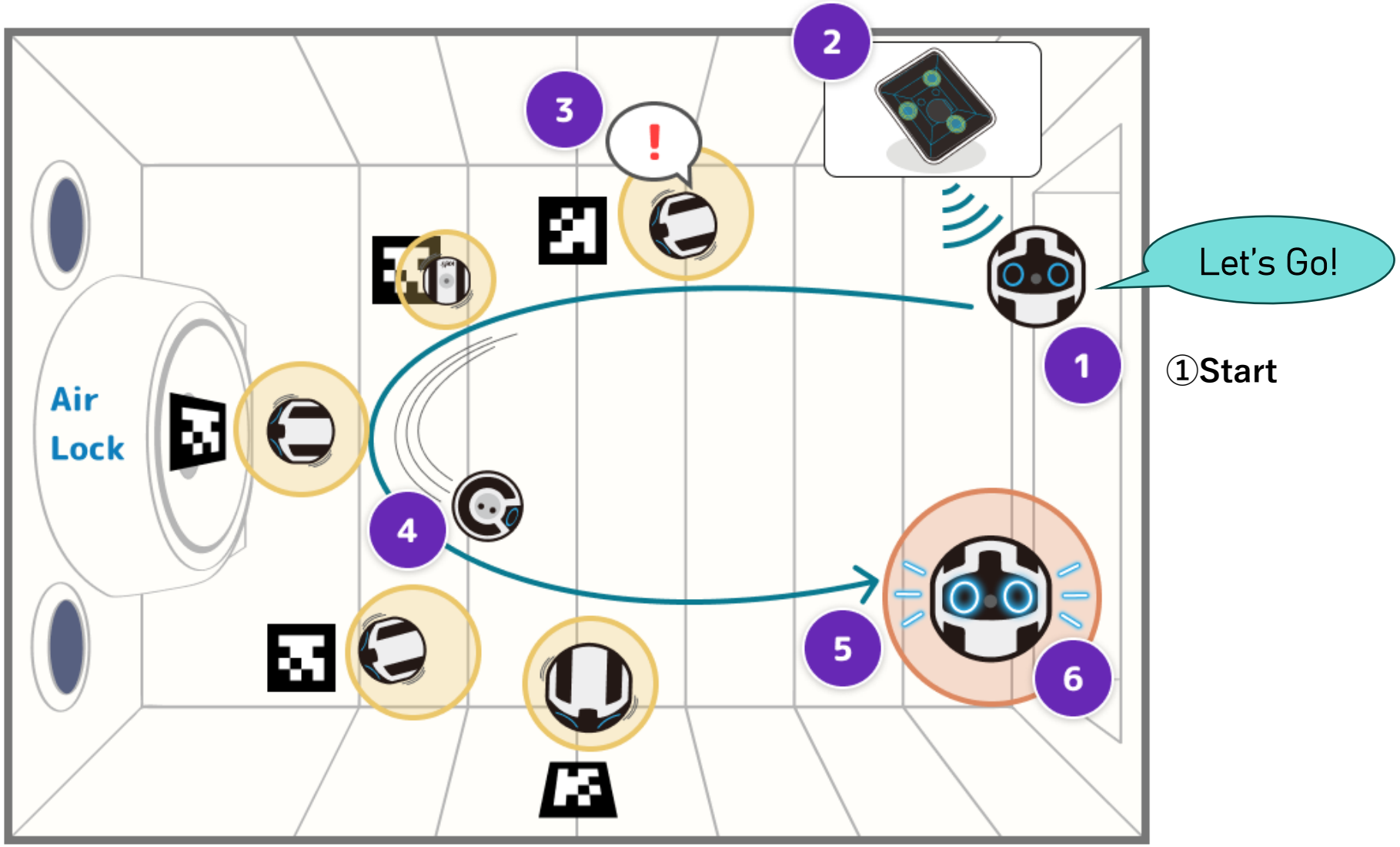
④ **Dock Station**: The platform in Kibo module where Int-Ball2 is charged and kept on standby. It is the start position of Int-Ball2. Related terms:

⑤ **Dock**: Secure Int-Ball2 to the Dock Station

⑥ **Undock:** Release Int-Ball2 from the Dock Station. Int-Ball2 is equipped with autonomous docking and battery recharging capabilities.

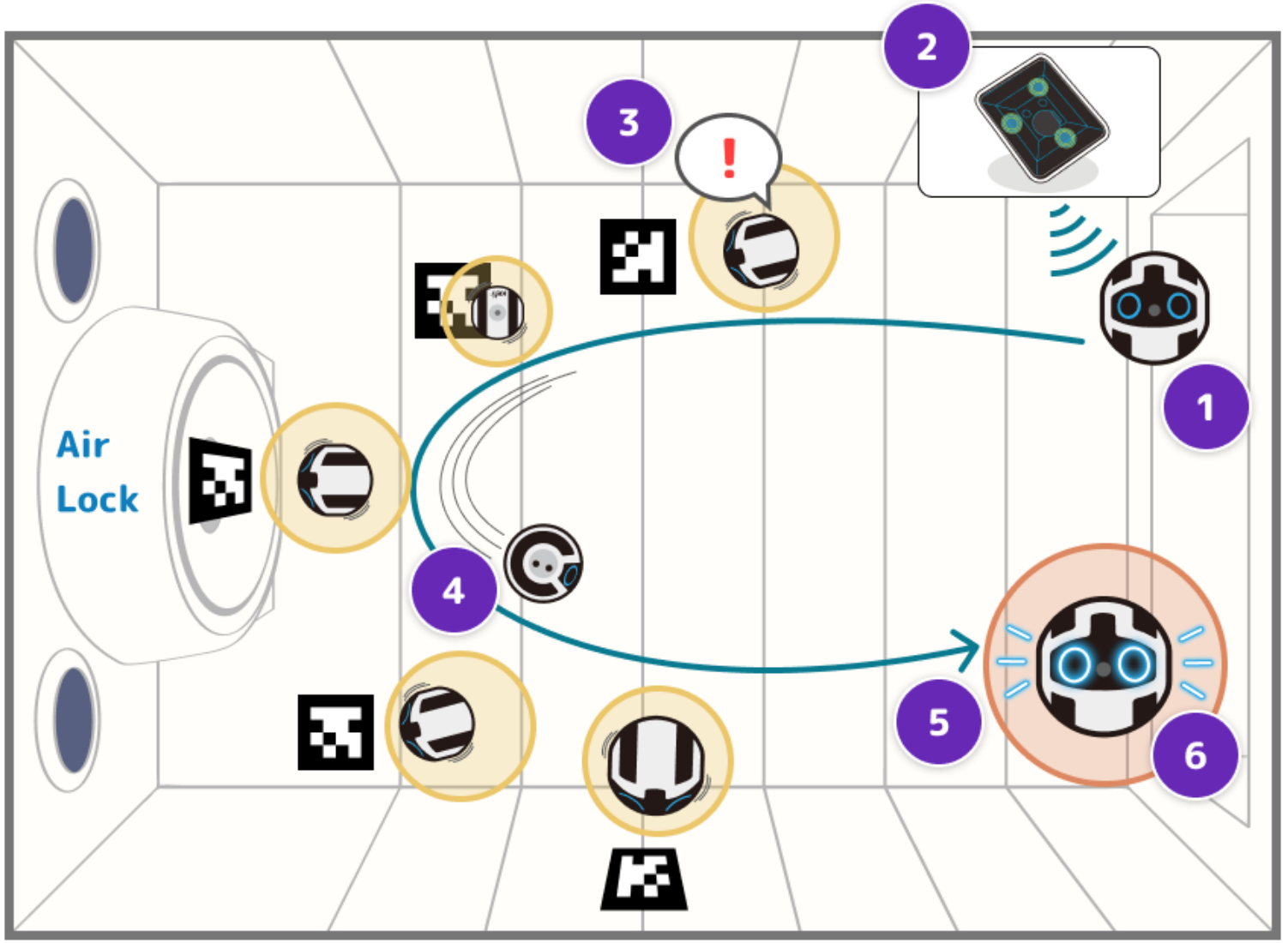
OUTLINE OF THE 7TH KIBO-RPC

- Overview of the Game (Mission)



OUTLINE OF THE 7TH KIBO-RPC

- Overview of the Game (Mission)

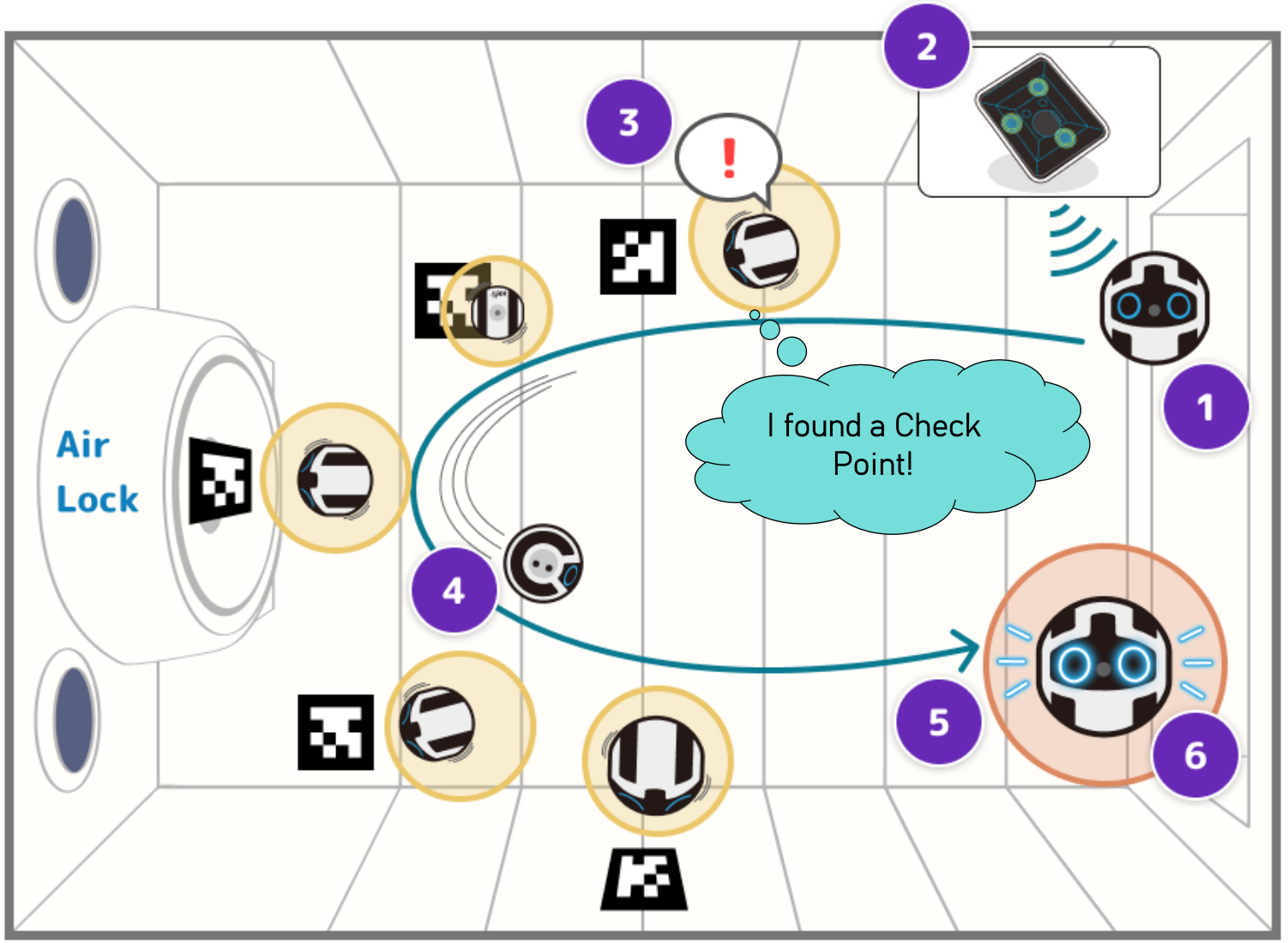


Read the automatically provided position and attitude information of the Check Points.

OUTLINE OF THE 7TH KIBO-RPC

- Overview of the Game (Mission)

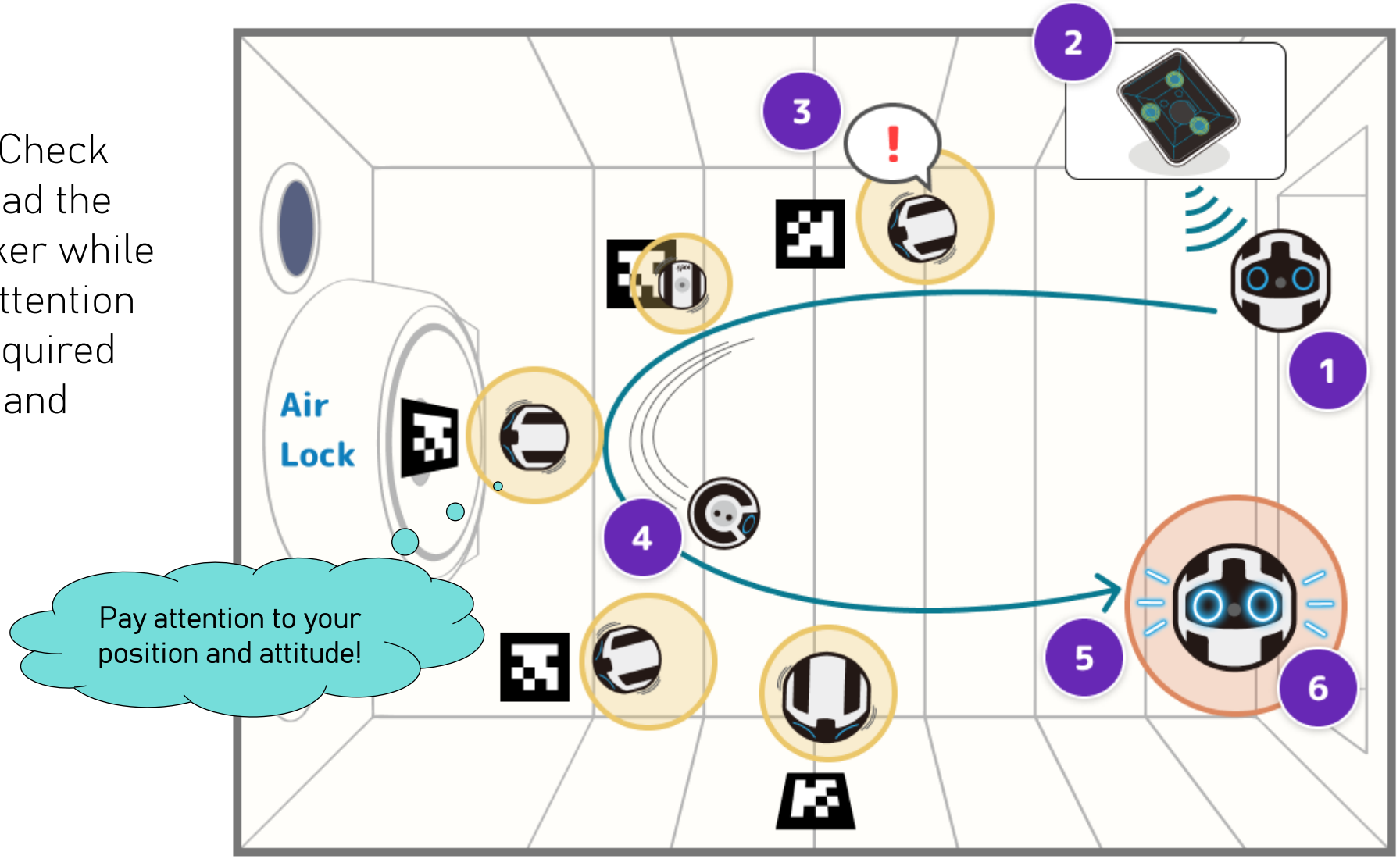
Plan an efficient route and visit the provided Check Points.



OUTLINE OF THE 7TH KIBO-RPC

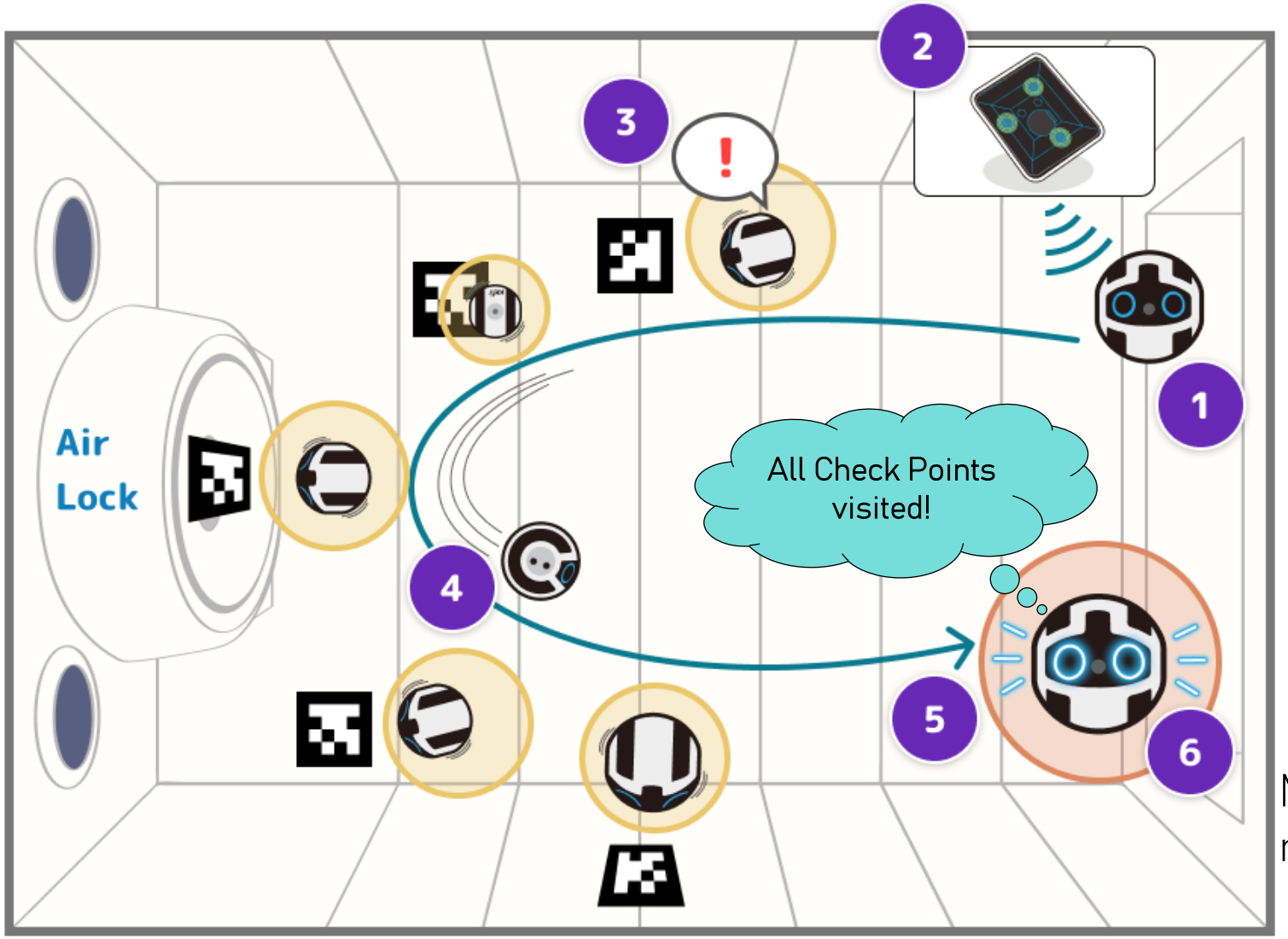
- Overview of the Game (Mission)

At each Check Point, read the AR marker while paying attention to the required position and attitude.



OUTLINE OF THE 7TH KIBO-RPC

- Overview of the Game (Mission)



Visited all Check Points and reached the Goal Check Point

Mission completion report is submitted

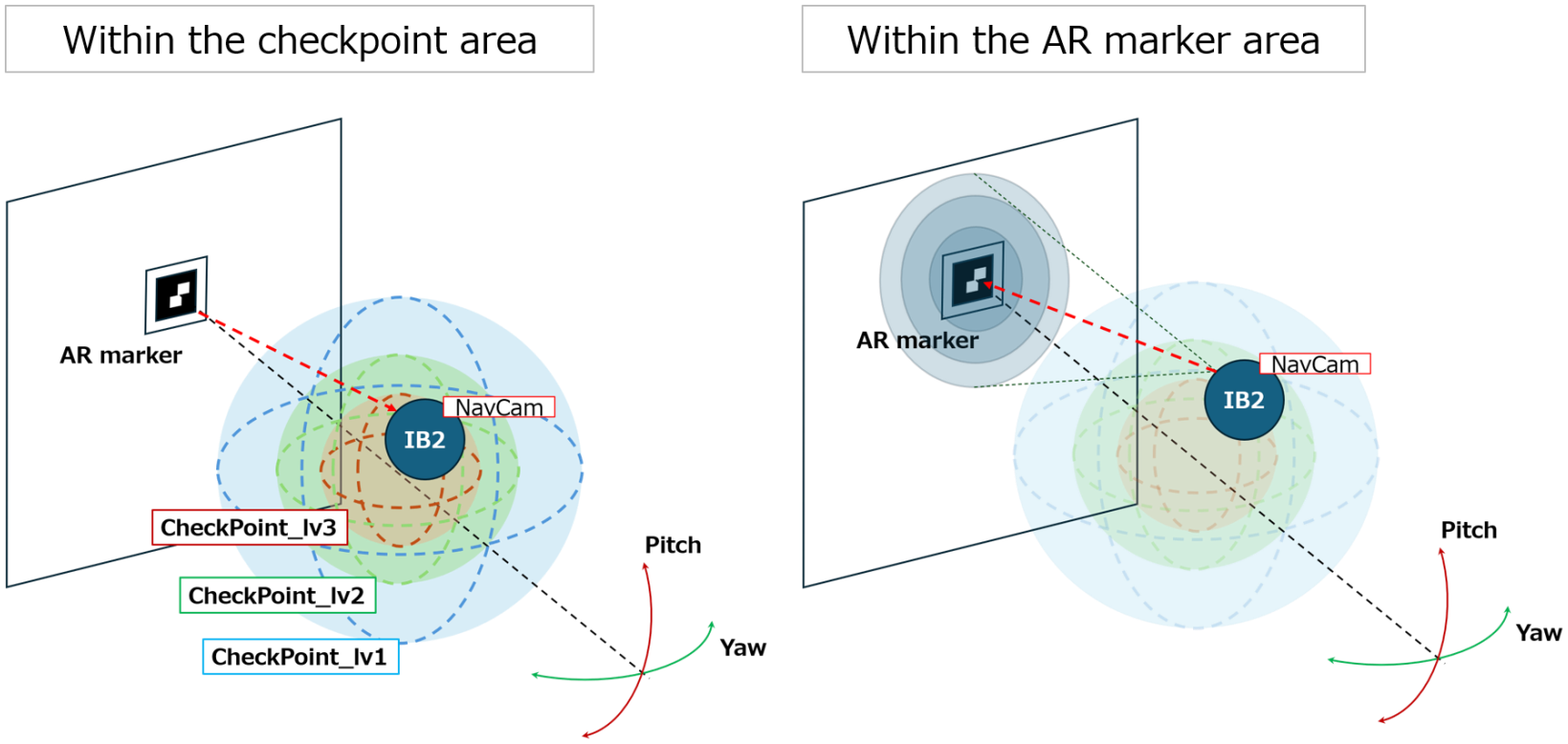
OUTLINE OF THE 7TH KIBO-RPC

Scoring Factors

1) Check Point Passing Judgment

Points are awarded when all three conditions are satisfied and this state is maintained for 1 second:
① the position of Int-Ball2,
② the AR marker is captured at the center of the camera, and
③ the attitude about the camera line-of-sight axis.

Example of Points Awarded



OUTLINE OF THE 7TH KIBO-RPC

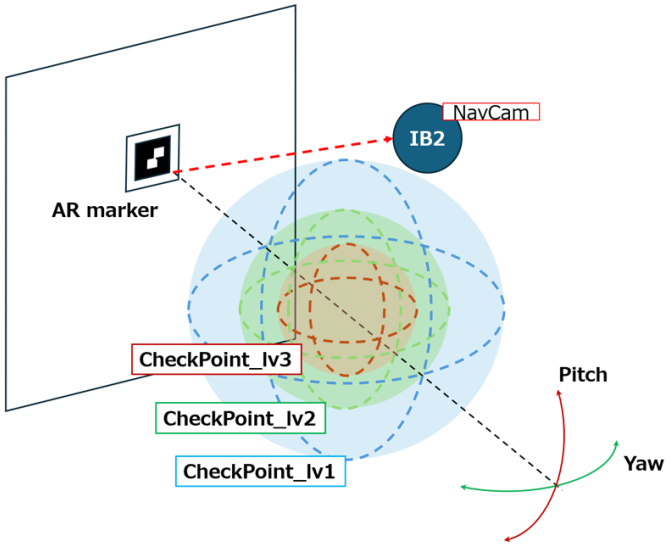
Scoring Factors

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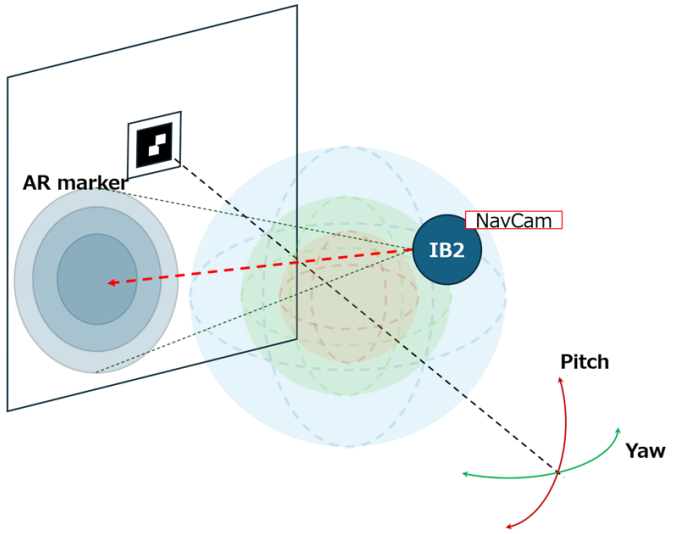
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① the position of Int-Ball2,
② the AR marker is captured at the center of the camera, and
③ the attitude about the camera line-of-sight axis.

Example of No Points Awarded

Outside the checkpoint area



Outside the AR marker area



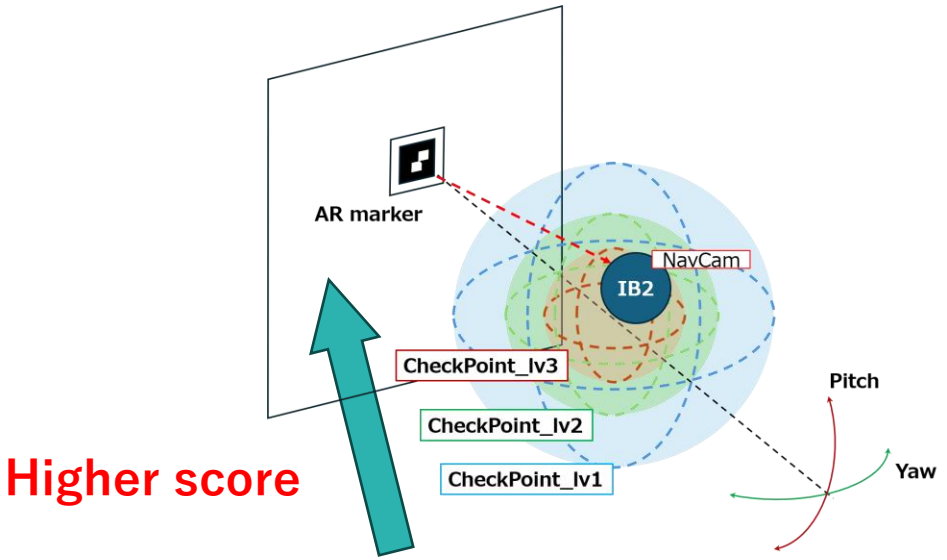
OUTLINE OF THE 7TH KIBO-RPC

A three-level margin, Lv1 to Lv3, is set for each Check Point.

The points you earn depend on the level at which Int-Ball2 reads the AR marker.

- **Lv1: Easy-to-read area (base points)**
- **Lv2: Slightly difficult area (additional points)**
- **Lv3: Most difficult area (highest points)**

The more accurately you read the AR marker from a more difficult position, the more points you earn.

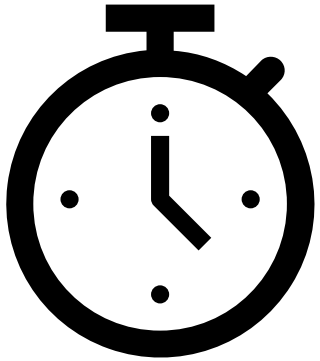


OUTLINE OF THE 7TH KIBO-RPC

Scoring Factors

2) Mission Time Remaining

The sooner you complete the mission, the higher your multiplier will be. A multiplier based on the time remaining is applied to your total score.



More time remaining
= higher multiplier

OUTLINE OF THE 7TH KIBO-RPC

Tips

- In this mission, it is important to be able to accurately scan the AR marker with relative position and attitude!
  - ✓ The key is to have Int-Ball2 accurately recognize the AR markers and the level of your relative position!
- Consideration of the strategic route design may lead to an increase in scores!
  - ✓ Whether you prioritize accuracy or choose a route that focuses on speed, design your route strategically to aim for a high score!
- Think about various recovery methods and keep them in mind in case things do not go as expected! (for example: it goes on a loop, markers aren't being loaded, the mapping is insufficient etc.)
  - ✓ It is essential to know how many errors to anticipate and how well prepared you are to deal with them.
- Ultimately, a team that takes the time to run many simulations will be the stronger team!

SCHEDULE



Open Simulator

Issue simulator login accounts
Issue Rulebook, Programming Manual

Preliminary Round

Program modification deadline

Final Round Event

@ JAXA Tsukuba Space Center

ISS Final Round Run

Guidance Session

Overall

7/6 • 7/16 • 10/15 • 11/4 • Jan 21 – Feb 17, 2027 • Mid-Apr. • *May* • *June*

Students

Send application

by Nov 4, 23:59 (JST)

Development period

Oct 2026 – Jan 20, 2027

Teams that applied develop programs using the simulator

Self-study

Finalists

Modify Program

Modify the program for the ISS Final Round Run

Finalists

ISS Final Round Run

Watch your run via Zoom

Teams that submitted a program

Participate in the Preliminary Round

Teams that applied can join the same-day workshop online.

Finalists

Join the Final Round Event

JAXA Tsukuba Space Center / online

Participants can keep developing their programs from Simulator opening until the submission deadline!

PROCESS AFTER APPLICATION

Preparation

** Details are in the Programming Manual*

1. **Install WSL2 and Ubuntu (Windows 11 users only)**
   - Enable WSL2 and install Ubuntu
   - Confirm that Ubuntu is running on WSL2

Programming languages: C++, Python

2. **Install Docker**
   - Install Docker Engine
   - Confirm that it works correctly

3. **Install recommended tools (Git and VSCode)***
   - Install Git
   - Install Visual Studio Code (VSCode)
   - Add the WSL extension to VSCode

   * Not required, but recommended to make development smoother.

PROCESS AFTER APPLICATION

Once preparation is complete

1. **Obtain the development resources**
   - Download K-RPC 7 Simulator Resources
   - Extract the development resources
   - Set up the local environment if needed

Start from the development resources (KRPC7 Simulator Resources) provided on the Kibo-RPC official website

2. **Create your program, try simulations, debug, and improve it**

Watch the development guide videos!

- Tutorial videos show how to log in to the Kibo-RPC website and how to create a program.
- If you are creating a program for the first time, please watch the videos first.

A must-watch for programming beginners!
Videos on this year's game rules, too!



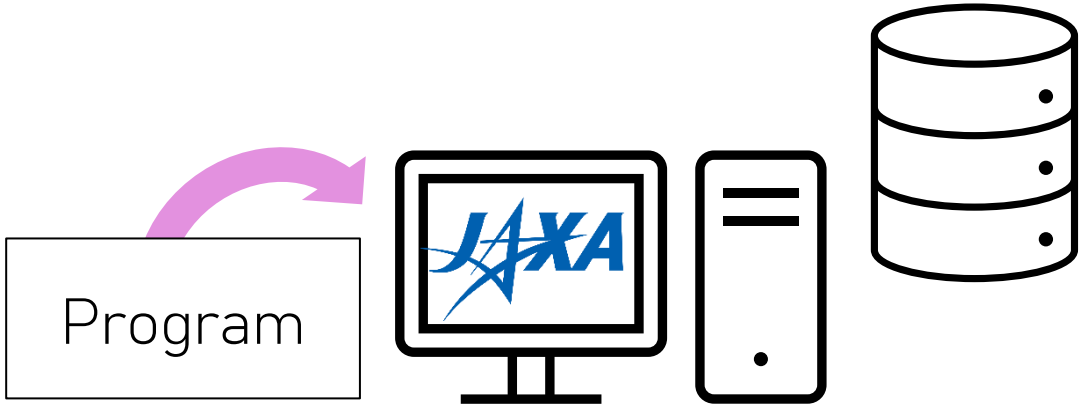
PROGRAM CREATION PROCESS

1. Program Creation



Create a program on your own PC
(C++, Python)

2. Upload the submission package to the Web Simulator



3. Run the simulation

PROGRAM CREATION PROCESS

Types of Simulators*

1. Web Simulator

   Used for pre-submission checks and evaluation in a common environment. Runs on a server managed by the Kibo-RPC organizers.

   It is a ARM64 environment, close to the actual hardware

   For details, please see the Programming Manual.

2. Local Simulator

   For verifying behavior and debugging during development. Runs on your own PC using Docker.

   Operating environment:

   - Ubuntu: 20.04 or later (24.04 recommended); Ubuntu on WSL2 for Windows 11
   - CPU: 4 cores or more
   - Memory: 8 GB or more
   - Storage: 20 GB or more of free space
   - Versions:

     GCC 7.5.0 / C++14 / Python 3.6.9 / ROS 1 Melodic (please do not update on your own)

* Perform basic checks on the Local Simulator and the final check on the Web Simulator.

If you cannot prepare a local environment, you can use the Web Simulator only (Docker is not required).

PROGRAM CREATION PROCESS

You may Log in to the Web Simulator with the account issued by the POC of your country or region.



https://jaxa.krpc.jp/

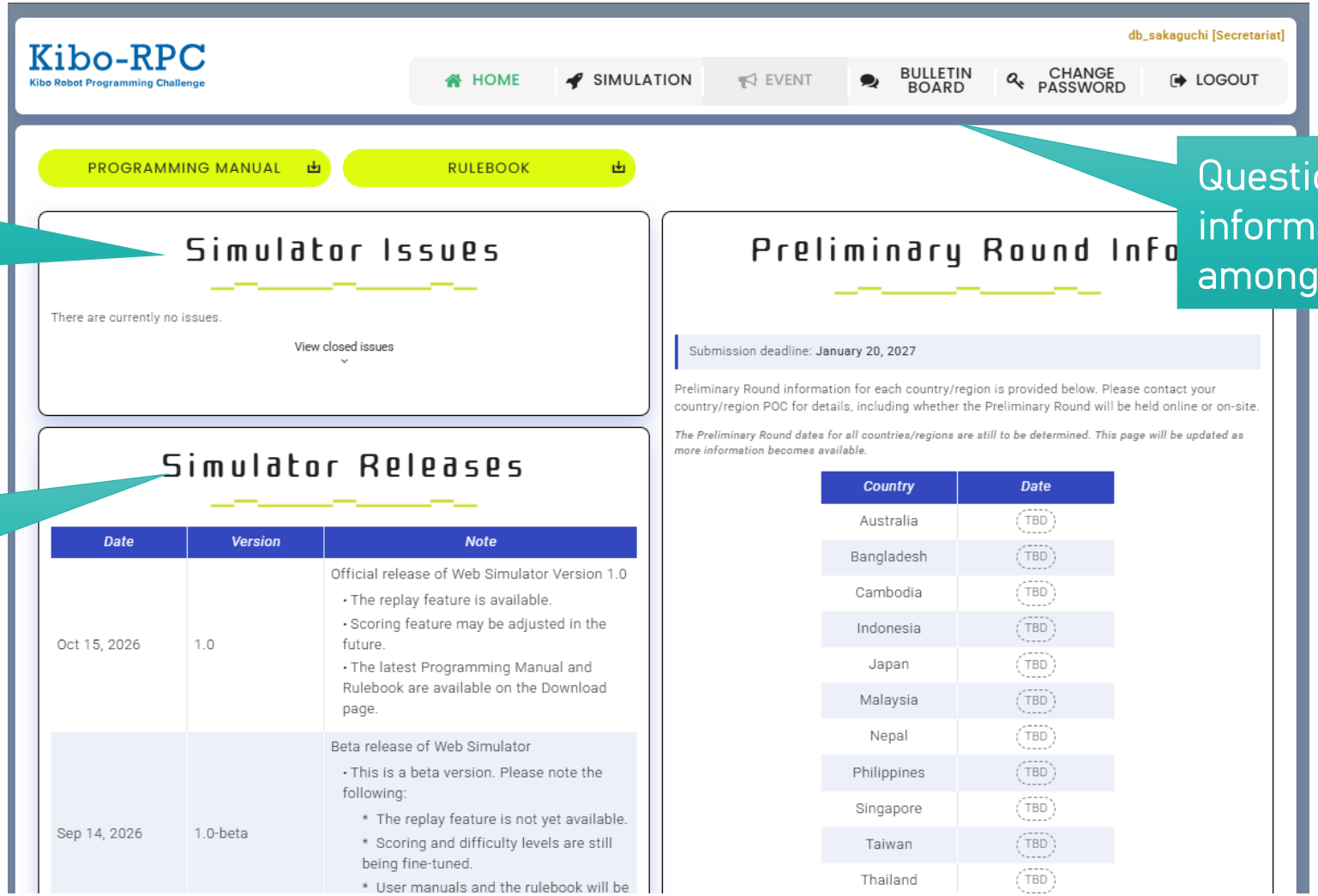
PROGRAM CREATION PROCESS

HOME screen

- Precautions
- Limitations
- Reported issues

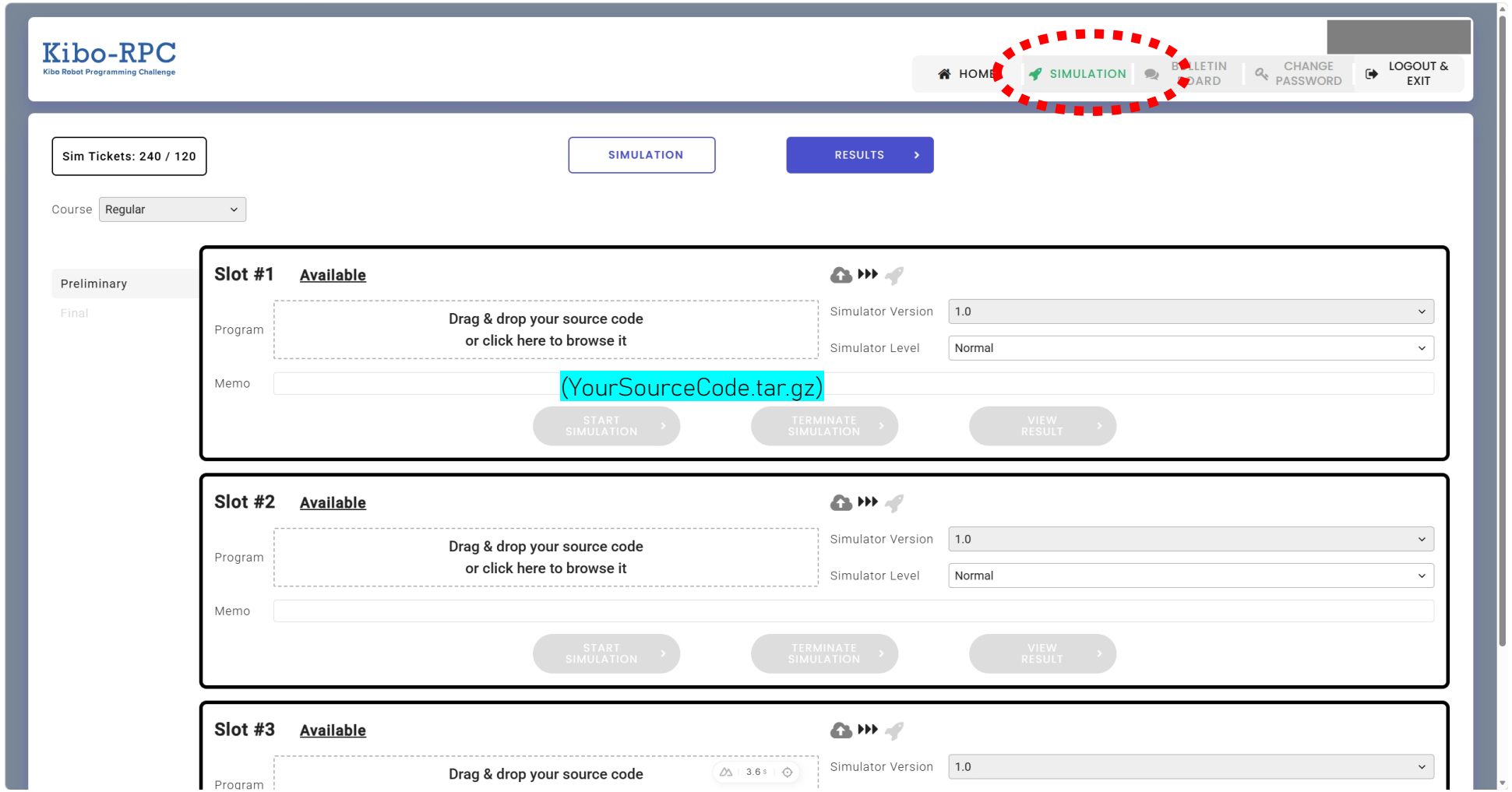
- What's New
- Version Number
- Release Date

Questions and information sharing among participants



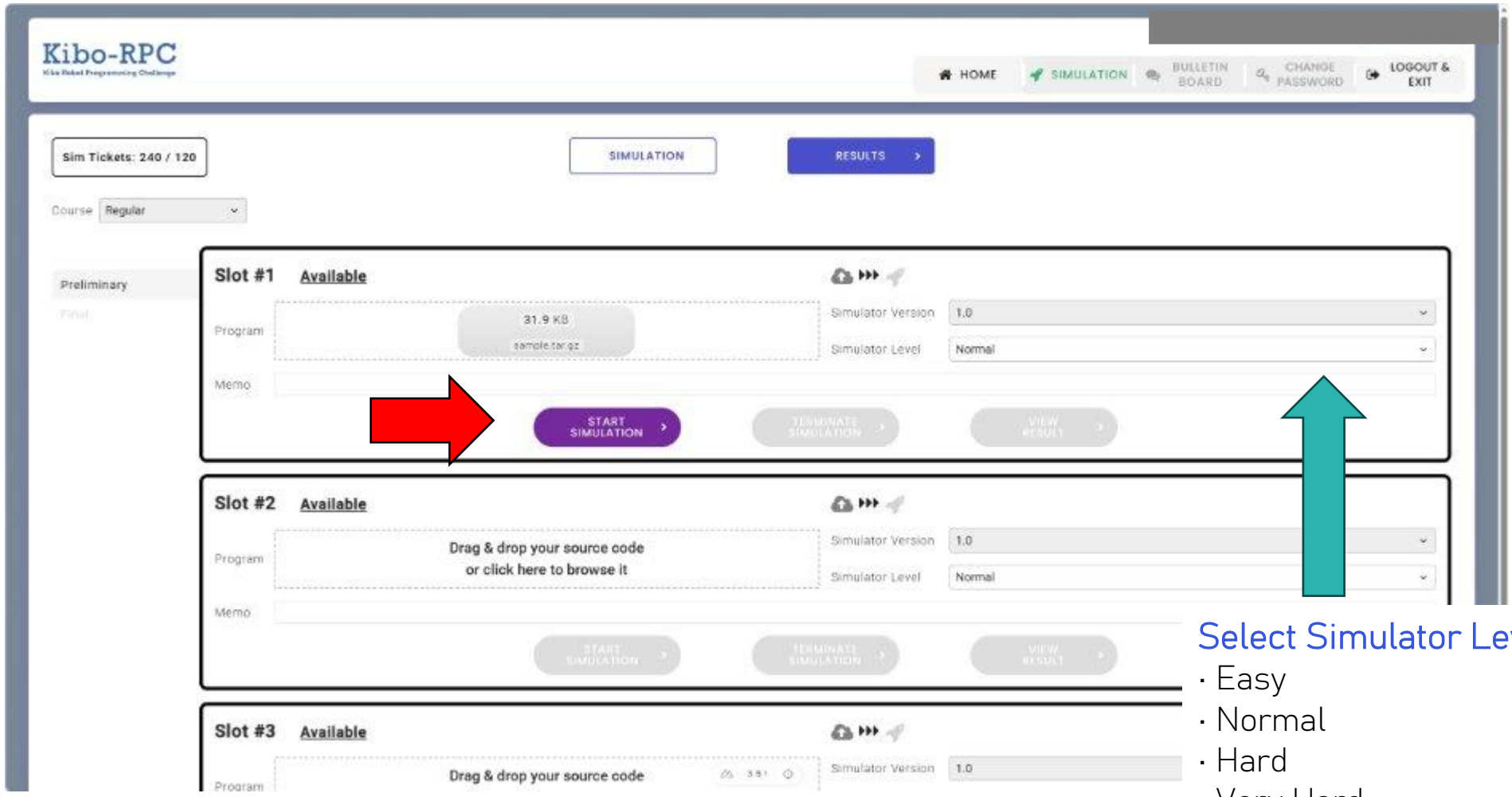
PROGRAM CREATION PROCESS

SIMULATION screen



PROGRAM CREATION PROCESS

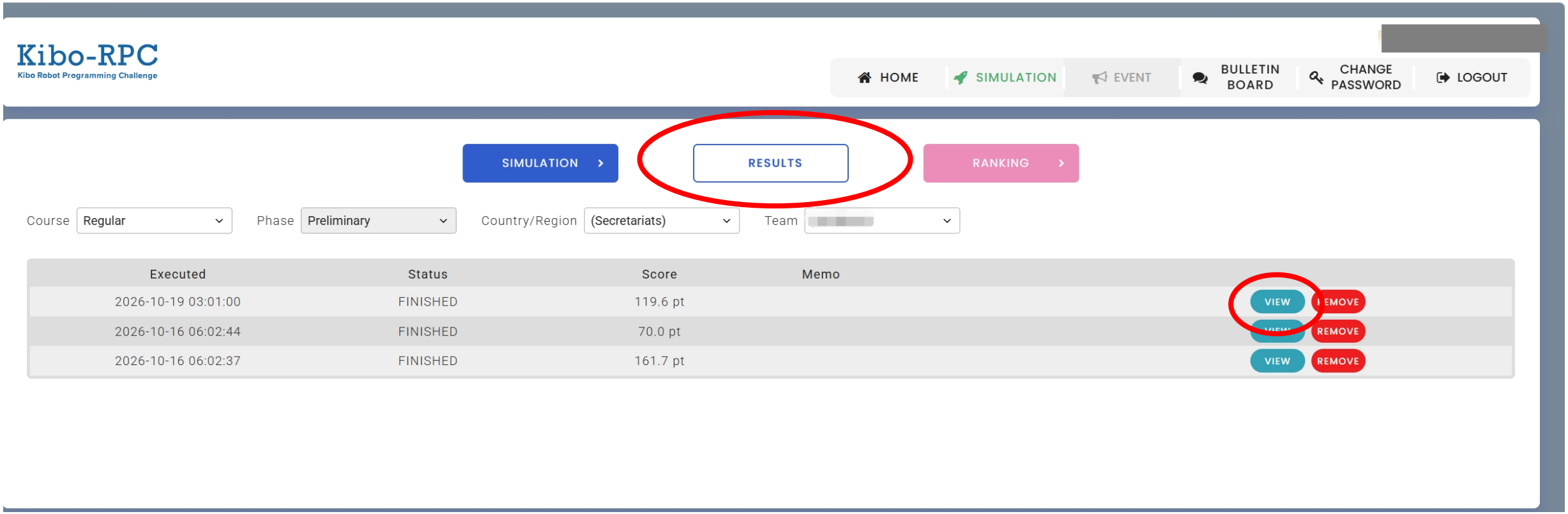
SIMULATION screen



Select Simulator Level

- Easy
- Normal
- Hard
- Very Hard

PROGRAM CREATION PROCESS



Up to 20 simulation results can be saved

When you exceed 20, delete previous results to save new ones.

PROGRAM EXECUTION PROCESS

BACK

Simulation Result

Status	Finished	Executed	2026-09-08 06:41:50
Team	[redacted]	Country/Region	[redacted]
Source Code	test_program_v2.tar.gz	DOWNLOAD SOURCE CODE	
Course	Regular	Simulator Version	
Simulator Level	Normal		
Memo			

Score Overview

Mission Result

Final Score	**106.70 pt**	Mission Time	**04:55.3**
Time Bonus	**x 1.02**	Goal Reached	Reached
CheckPoints Passed	**8 / 8**		

Score Breakdown

CheckPoint Score Total 75.00 pt

Level Summary

Level	Passed
Lv1	1
Lv2	0
Lv3	7

CheckPoint Summary

Goal Score 30.00 pt

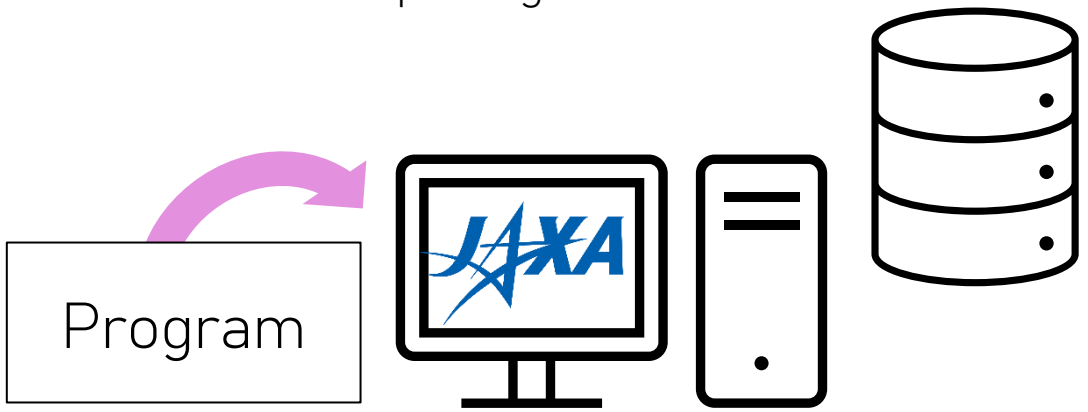
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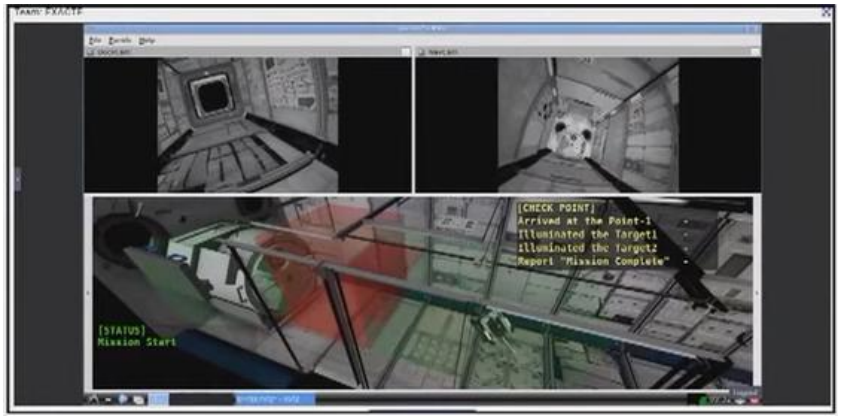


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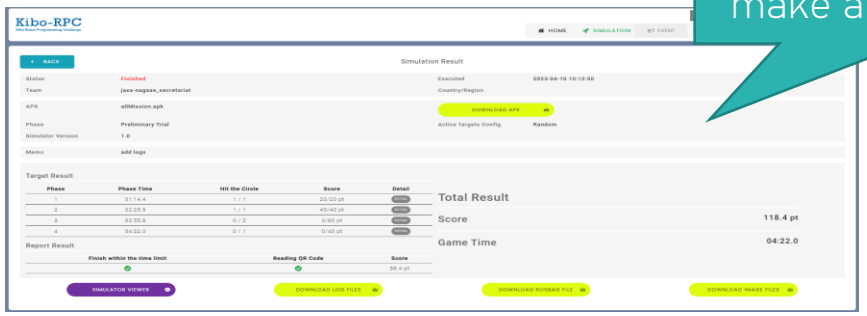
2. Upload the submission package to the Web Simulator



3. Run the simulation
(Up to 120 runs per month)



4. Results Verification



You can improve your program from the simulation results to make a better program!

TIPS FOR PROGRAMMING

Tips for creating stable programs

For details, please see the Programming Manual.

- **Avoid infinite loops**

  Set upper limits on loop counts and processing times, and limit the number of retries.

- **Error handling**

  Always check movement results and API return values. If something fails, log it and shut down safely.

- **Log design**

  Log the start and end of processing, API names and results, target/current position and attitude, error codes, and processing times.

- **Handling sensor data**

  Expect missing or stale sensor values, and state units and coordinate systems explicitly.

- **AR marker recognition**

  Use OpenCV (ArUco): 4×4 dictionary, marker IDs 1 to 12, with a 2 cm side length.

- **Common notes when using the API**

  Game API calls use ROS Services, Actions, and Topics (see the Programming Manual).

AGENDA

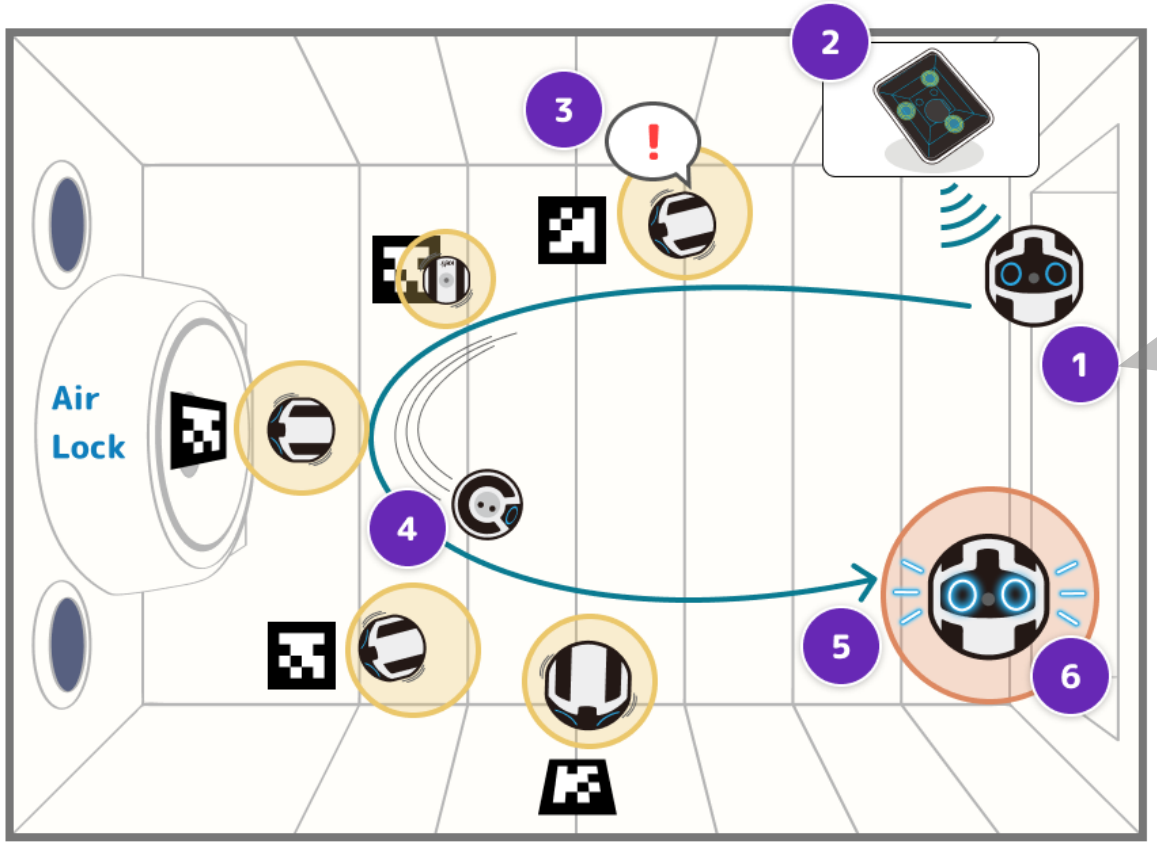
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ABOUT THE PRELIMINARY ROUND

Mission

- In the Preliminary Round, the programs submitted by the deadline are run on the simulator in each country/region, and the team with the highest average score advances to the ISS Final Round Run as a finalist.

* Depending on the country/region, factors other than the simulation score may also be evaluated.



6 to 12 Check Points

ABOUT THE PRELIMINARY ROUND

Preliminary Round Rules

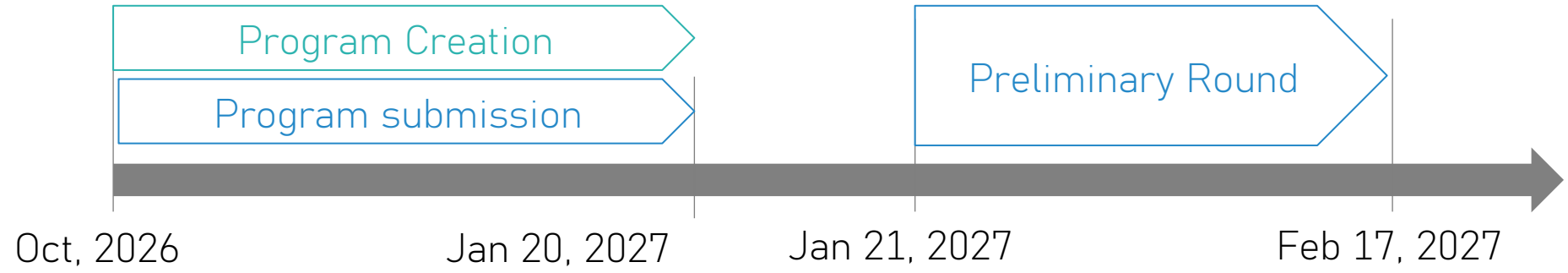
The details are included in the rulebook

1. Eligibility for Participation
   - Teams that have submitted the application form by 23:59 on Nov 4, 2026 (JST)
   - Teams that have submitted a program by the submission deadline

Program submission period: Oct 2026 to Jan 20, 2027

2. Preliminary round period

Preliminary round period: Jan 21 to Feb 17, 2027

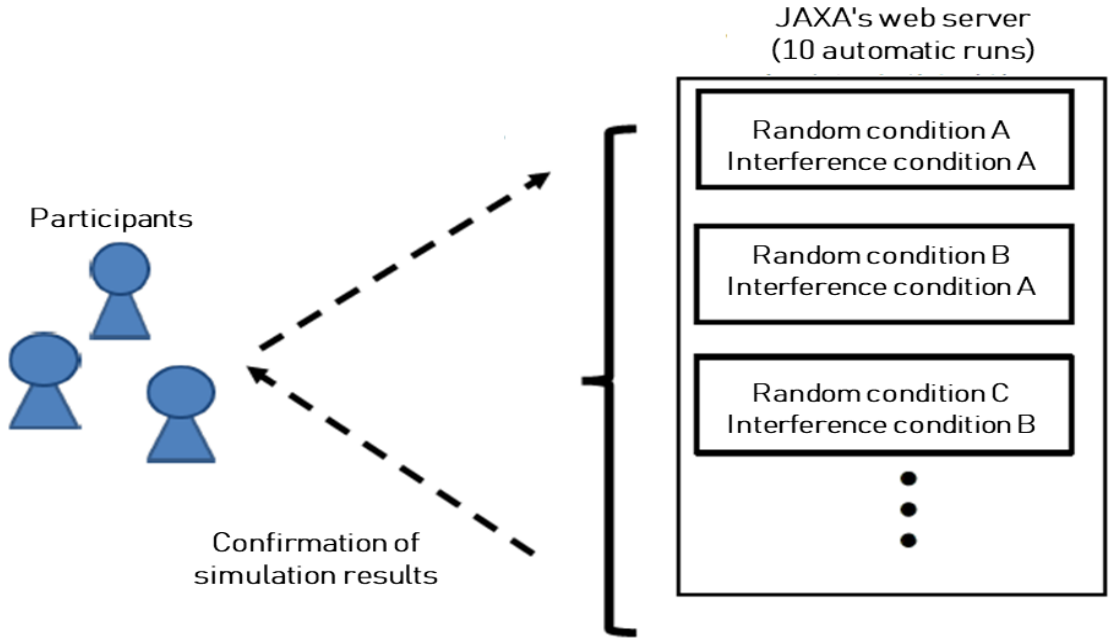


ABOUT THE PRELIMINARY ROUND

Preliminary Round Rules

1. Only one program per team can be used

2. 10 Runs for each team in the preliminary round
   - 10 Runs per program
   - Random conditions differ for each Run, but are the same for all teams
   - Score is calculated on the average of 10 Runs



ABOUT THE PRELIMINARY ROUND

Preliminary Round

January 21 to February 17 2027

The event will be held on one day over the above period, depending on local organization(s)/agencies in each country/region.

Please wait for further announcements from the relevant agencies in each country or region for details.





AGENDA

1. Overview of the Kibo Robot Programming Challenge (Kibo-RPC)
2. Outline of the 7th Kibo-RPC
3. Preliminary Round
4. **ISS Final Round Run**
5. Beginner Course
6. Q&A

ABOUT THE ISS FINAL ROUND RUN

- Mission Details
- The ISS Final Round Run will be conducted by uplinking (installing) the program onto Int-Ball2 on the ISS.
- The modified program for the ISS Final Round Run must be submitted in advance (scheduled for late Feb to mid-April 2027).
- Details regarding the ISS Final Round Run and the program submission deadline will be announced to the finalists at a later date.

ABOUT THE FINAL ROUND

The ISS Final Round Run and the Final Round Event will be held on different days.

ISS Final Round Run: Early May 2027

Final Round Event on the ground: June 2027

Final Round Event Details
Greetings from the organizer/Explanation of the rules
Introduction of participating teams and ISS run results
Announcement of standings/award ceremony/interview with the winning team
Kibo-RPC Workshop (Lecture and Networking)

*Details of 6th Kibo-RPC

The last event can be watched on YouTube
https://www.youtube.com/watch?v=fVhdATbsMJk







The 6th Kibo-RPC Final Round Event was held in a hybrid format online and at the Tsukuba Space Center

AGENDA

1. Overview of the Kibo Robot Programming Challenge (Kibo-RPC)
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ABOUT BEGINNER COURSE



- Overview

For the 7th Kibo-RPC, in addition to the traditional Regular Course, a Beginner Course will be offered.

In the Beginner Course, programs will be run on a simulator, similar to the Preliminary Round for the Regular Course.

Teams will compete for rankings under an easier game level compared to the Regular Course. The easier conditions consist of fewer Check Points, and the attitude is not considered.

The finalist teams from each country/region will advance to the final competition. Awards will be given to the top teams, but please note that even if you win, you will not be eligible to participate in the ISS Final Round Run aboard the ISS.

Teams that meet the following criteria are eligible to participate:

- All team members must be junior high school age or younger (at the time of submission).
- All team members are programming beginners (self-proclaimed)
- **Teams that meet the above two conditions and wish to participate.** All teams can also participate in the Regular Course.
- The program submission can be the same for all teams, but it must be submitted to each course separately.
- **The simulator can be run up to 120 times per month. If you run out of runs, please contact the secretariat.**

For the rules of the Beginner Course, please refer to the rulebook.

ABOUT BEGINNER COURSE

	Regular Course	Beginner Course
Participants	**Elementary school to graduate school students** as of submission deadline for the Preliminary Round	**Junior high school students** and under as of submission deadline for the Preliminary Round
Difficulty	Standard	Easy
Application	Apply for this course only	Can also apply for the Regular Course
Experience	No restriction	All members are beginners in coding
Number of participants	3-8	3-8

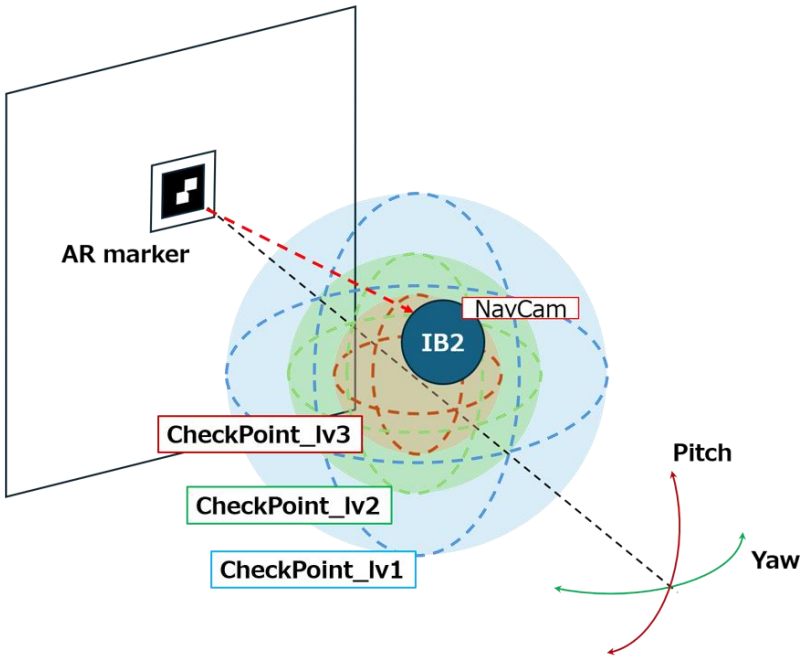
ABOUT BEGINNER COURSE

Scoring Factors

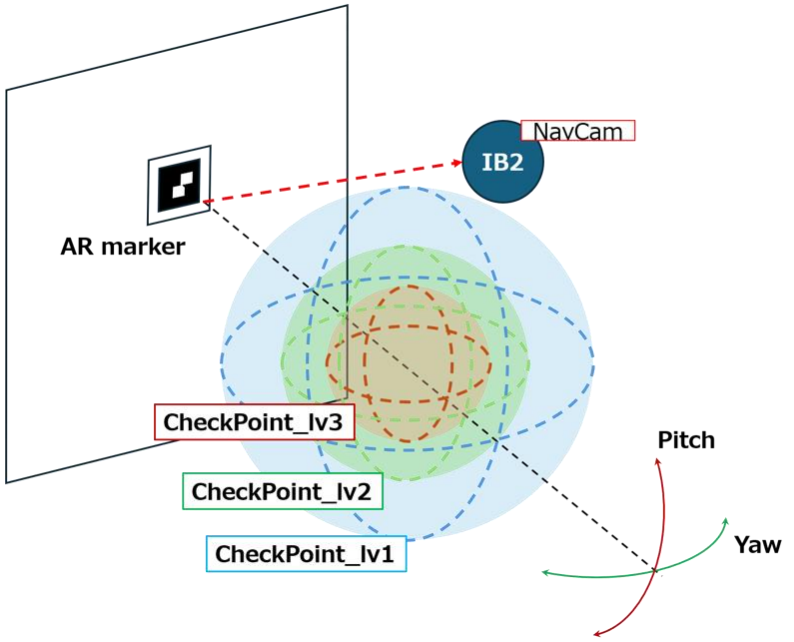
1) Check Point Passing Judgment

Points are awarded when the position of Int-Ball2 is within the specified margin and stays there for 1 second.
Unlike the Regular Course, attitude control and AR marker reading are not scored.

Example of Points Awarded



Example of No Points Awarded



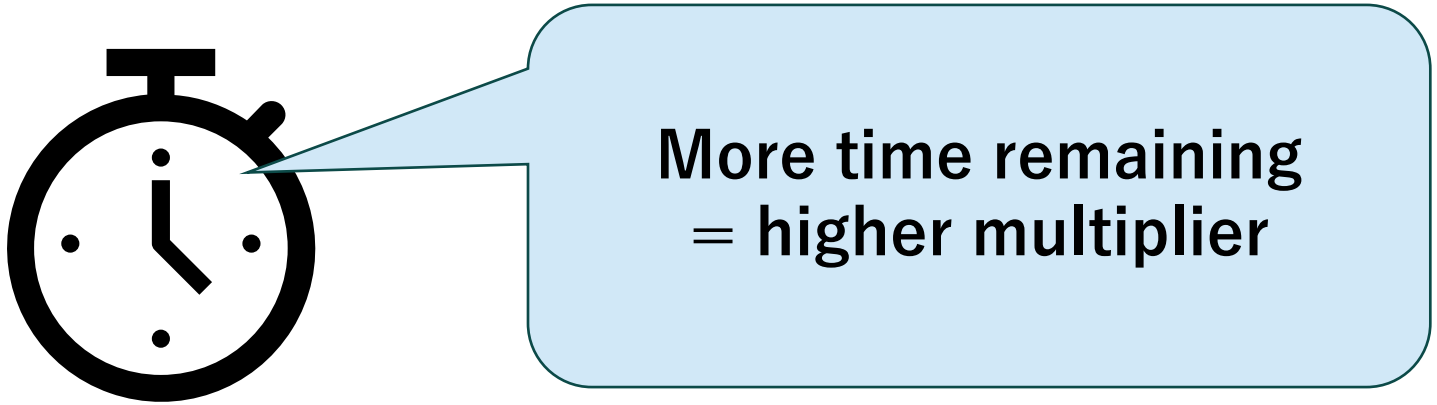
ABOUT BEGINNER COURSE

Scoring Factors

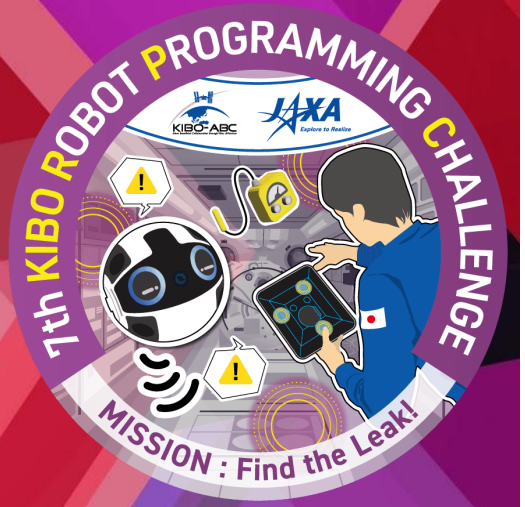
2) Mission Time Remaining (same as the Regular Course)

The sooner you complete the mission, the higher your multiplier will be.

A multiplier based on the time remaining is applied to your total score.



THE 7TH KIBO ROBOT PROGRAMMING CHALLENGE GUIDANCE SESSION



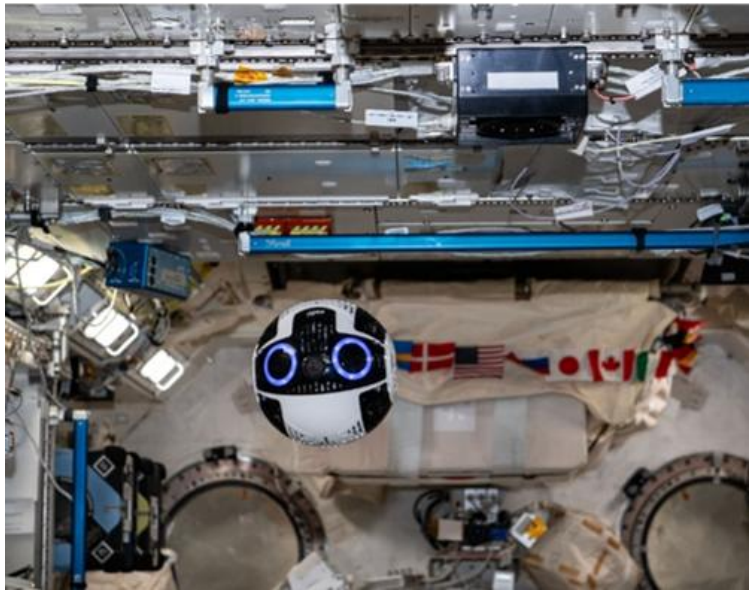
Thank you for your attention!

We look forward to your participation!

Q&A

- If you have any questions, please ask them in the Chat.

*Questions that require confirmation will be answered in the FAQ on the Kibo-RPC Official Site.



7th Kibo-RPC Competition URL: https://jaxa.krpc.jp/
JAXA KUOA URL: https://humans-in-space.jaxa.jp/biz-lab/kuoa/
Kibo-RPC Secretariat: z-krpc7@ml.jaxa.jp